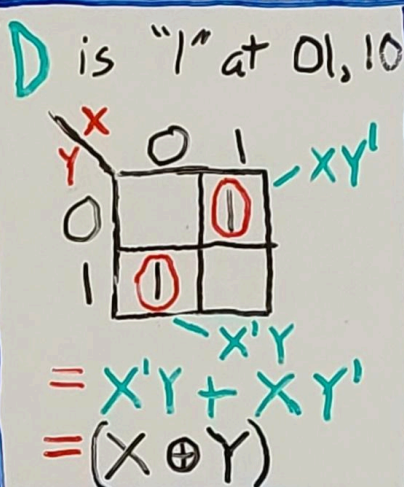
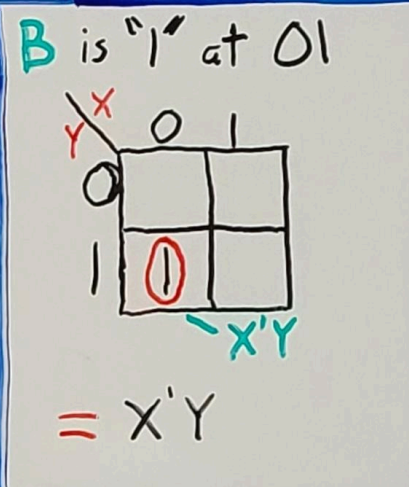


Lab Report #5

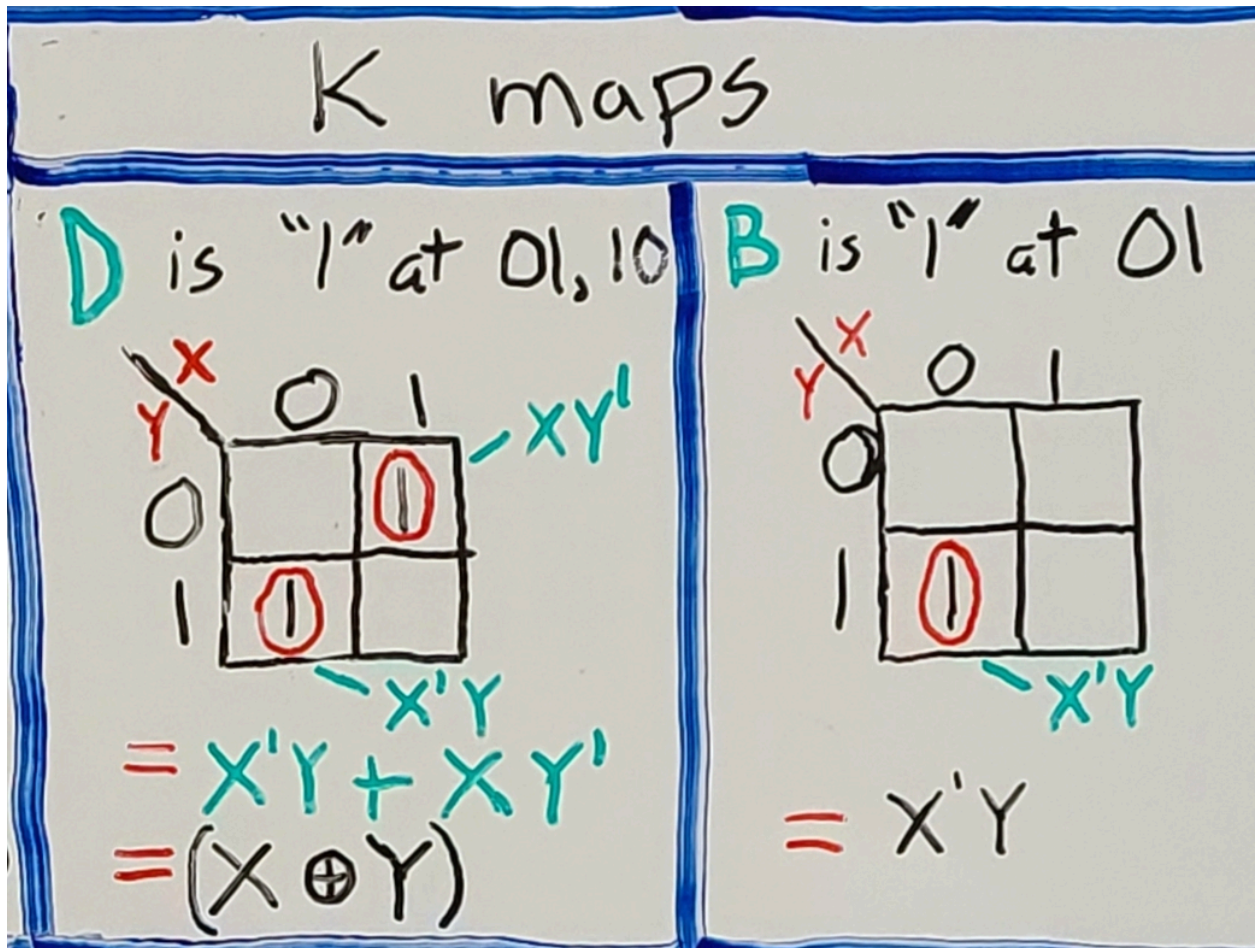
Section 5.1

1) Truth Table

Lab 5: Half Subtractor			
Truth Table		K maps	
X	Y	D	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

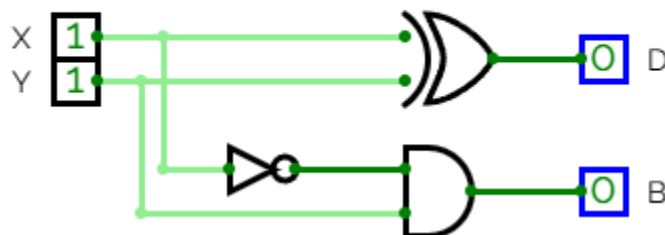
D is "1" at 01, 10  $D = X'Y + XY'$ $D = (X \oplus Y)$		B is "1" at 01  $B = X'Y$	
---	--	--	--

2) B and D in terms of x and y variables



3) Circuit Diagram

Half-Subtractor



<https://circuitverse.org/users/403924/projects/lab-5-group-2>

4) Observations

Our circuit was observed to match our expected truth table values.

Section 5.2

1) Truth Table

Lab 5: Full Subtractor

Truth Table			K maps
X	Y	Z	D B
0	0	0	0 0
0	0	1	1 1
0	1	0	1 1
0	1	1	0 1
1	0	0	1 0
1	0	1	0 0
1	1	0	0 0
1	1	1	1 1

D is "1" at 001, 010, 100, 111

B is "1" at 001, 010, 011, 111

D K-map:

	00	01	11	10
0		1		1
1	1		1	

$$= XYZ + XY'Z' + X'YZ' + X'Y'Z$$

$$= (X \oplus Y) \oplus Z$$

B K-map:

	00	01	11	10
0		1		
1	1	1	1	

$$= X'Z + YZ + X'Y$$

2) K-maps

K maps

D is "1" at 001, 010, 100, 111

B is "1" at 001, 010, 011, 111

D K-map:

	00	01	11	10
0		1		1
1	1		1	

$$= XYZ + XY'Z' + X'YZ' + X'Y'Z$$

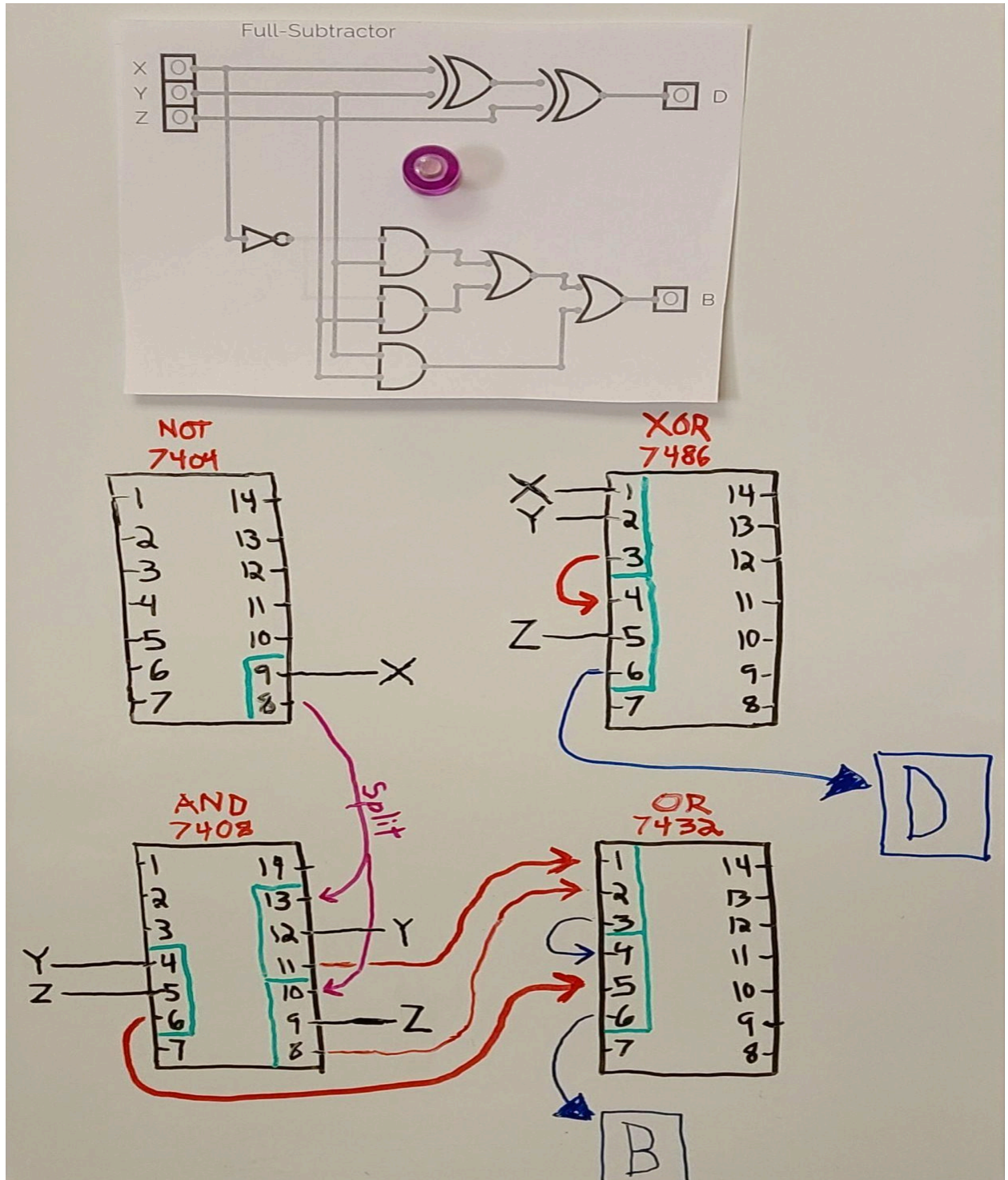
$$= (X \oplus Y) \oplus Z$$

B K-map:

	00	01	11	10
0		1		
1	1	1	1	

$$= X'Z + YZ + X'Y$$

- 3) Expression of functions D and B in terms of x,y,z
 $D = X \oplus Y \oplus Z = XYZ + XY'Z' + X'YZ' + X'Y'Z$
 $B = X'Z + YZ + X'Y$
- 4) Circuit Diagram of Full-subtractor



5) Prove that functions D and B 4) are equal to 3)

$$\begin{aligned}
 & (x \oplus y) \oplus z \\
 & x \oplus y = xy' + x'y \\
 & (xy' + x'y) \oplus z \\
 & (xy' + x'y)z' + (xy' + x'y)z \\
 & xy'z' + x'yz' + (xy')'(x'y)'z \\
 & \quad \text{De Morgan} \\
 & \quad (x'+y)(x+y')z \\
 & \quad \cancel{x'x} + \cancel{x'y'} + yx + \cancel{yy'} \\
 & \quad (x'y' + yx)z \\
 & \quad x'y'z + yxz \\
 & \boxed{xy'z' + x'yz' + x'y'z + yxz} \\
 & \text{Q.E.D.}
 \end{aligned}$$

6) Observations

Our circuit was observed to match the expected truth table values for D and B